

LatticeRank Phase 2 — Failure Analysis

All numbers below are measured on our own generator (data/phase2/p2_val + p2_stress), which is harder than the disclosed blind-set spec. We measured why rather than assuming: the 40% decoy rate and the severity ladder — the two things it was built to stress — have **no measurable effect** on localization (§1b). What does is that we render the reference and search as independent acquisitions, worth **35 points of ≤1 px rate**. Diagnostic numbers, not predictions for Set A.

1. Diagnosis: localization error is binary; selection is the bottleneck

At n=180 the pairs landing within 1, 2, 3 and 5 px are the **same 33 pairs** — none fell between 1 and 5 px. A correct site lands sub-pixel; a wrong one misses by hundreds. **Mean tiered credit therefore equals site-selection accuracy**; subpixel refinement cannot move this score block.

Stage	Rate	Note
Detection (true site is a local max)	~99–100%	not the problem
Pool (true site survives harvest, GT pose)	90–95%	not the problem
Selection (ranker picks it)	15–43%	the bottleneck

Evidence: results/gt_pose_ceiling.json (n=316); PHASE2_FINDINGS.md §3, §5.

Pool-high, selection-poor is an *identity-evidence* gap, not recall/cap/NMS. This sharpens Phase 1's own flagged weakness — internal selection at 48.75% against a 90% pool (README.md) — Phase 2 shows the same shape persisted under unknown pose: pool recall held at 90–95% while selection fell to 15–43%. Root cause, measured directly on 67,370 held-out candidates: the current lattice-relative features (disp_periods x/y, lat_phase_res, parity_even) score **AUC 0.38–0.62** against the true-site label — near chance (results/phys_eval_full.json). Seven hand-designed selectors and two retrained rankers all land in the same 0.18–0.28 band for the same reason: re-weighting existing evidence cannot fix a feature set that does not carry the signal.

1b. Why our corpus is hard — measured, not assumed

Set A is specified as “noise comparable to the Phase 1 sample prompt” and decoys are never mentioned, so the worry was that we had made the task artificially hard. Three arms of the shipped pipeline, no code changes: severity 0–1 with **no decoys** (n=180) gives **30.0% ≤1 px** [23.8, 37.1]; severity 0–1 **with decoys** (n=110) gives **31.8%** [23.9, 41.0]; the full severity mix (n=110) gives **30.0%** [22.2, 39.1]. Indistinguishable — **neither decoys nor severity move localization**, so Set A conditions should not be expected to rescue the block.

What does move it is a generator design choice. We render the reference and the search as **independent acquisitions** of one latent scene — separate RNG streams, so the noise realizations differ. Generators that instead crop the reference from the rendered search pixels leave the noise pattern as a unique fingerprint of the true site. Paired, n=60, identical pairs, only the reference construction differing:

arm	≤1 px	95% CI	pts/40	median err
independent acquisition (ours)	30.0%	[19.9, 42.5]	12.9	203.62 px
reference cropped from search pixels	65.0%	[52.4, 75.8]	33.5	0.87 px

Exact McNemar 26 vs 5, **p = 1.9e-4**, **Δ +35.0 pp**. By severity: +11.8 / +33.3 / +18.2 / **+90.9 pp** at levels 0–3 — at severity 3 the pipeline goes 1/11 → 11/11. That is the noise-fingerprint signature: severe noise is catastrophic when independent and a strong unique cue when shared. This is an **upper bound, not a prediction** — a real same-canvas generator still applies its own reference-side PSF and noise, so an unknown generator lands inside [12.9, 33.5]/40. The point stands regardless: the 20-point spread on the largest scored block is set by a generator choice we cannot observe, not by anything reachable in the solver. The pipeline is sub-pixel (0.87 px median) the moment a shared-noise cue exists.

1c. The ceiling, measured with an oracle

The decisive test is not “can we rank better” but “is the true site separable at all”. Hand the matcher the **ground-truth pose** and remove search from the problem entirely (localization_ceiling.json n=80; dense_residual_ceiling.json n=60):

arm	≤1 px	95% CI
shipped (global argmax over 45 poses)	25.0%	[16.8, 35.5]
oracle: ground-truth pose, band-passed ZNCC	31.2%	[22.2, 42.1]
oracle: ground-truth pose, dense periodic residual	36.7%	[25.6, 49.3]

Perfect pose knowledge is worth **~6 pp, not 70**. At the true pose the true site is the global maximum in only **14–16 of 60–80** pairs, with a median of 10–22 locations scoring higher. Dense residual matching — periodic cancellation, the mechanism at the heart of the Phase 1 method — is statistically indistinguishable from band-passed ZNCC (paired McNemar 5 v 4, **p = 1.00**). Two consequences follow. **(1) Pose search is not the bottleneck**, so grid refinement cannot help — consistent with the earlier null at 45→99→187 poses. **(2) Any rule computed over these surfaces is bounded above by ~35%**, because ranking cannot promote a site the similarity places 20th; candidate ranking, feature re-weighting and threshold tuning are all such rules, which is why ten families of them returned the same 0.18–0.28 band. The cross-pose information the sweep discards was tested and is *not* usable: summing across poses scores 2.5%, votes 1.2%, peakiness 18.8%, against 25.0% shipped. Within the extension space the addendum allows, **the localization block is capped near 13/40, and the cap is a property of the evidence rather than of the estimator**.

Seven targeted attacks. The loss splits into disjoint buckets at the estimated pose: 22.5% where the truth is not among the candidates at all, 45.0% where it is a candidate but is not picked. Each was attacked directly and all seven failed: oracle pose (+6 pp), cross-pose sum/votes/peakiness (2.5/1.2/18.8% against 25.0%), dense periodic residual (p=1.00), five ranking rules on identical shortlists (all below peak at 35.0%), NMS widening (structurally cannot help — the global argmax is NMS-independent), and four trimmed/robust similarities. The last is the informative negative: trimming degrades accuracy *monotonically* (35.0→33.3→31.7→30.0% at 0/10/25/40% dropped). If damage were localised, discarding the worst-agreeing pixels would rescue the true site; that it never does means the disagreement is spread uniformly across the template — the signature of independent acquisition noise, not occlusion, reached from the opposite direction to §1b. Behind the 22.5% bucket the truth's median percentile on the surface is **80.2%**: a fifth of the surface outscores it, so the defect is the true site's *score*, not its rank — which is why every re-ranking attack was bound to fail. Two further attacks close the remaining leads. The **DoG band** was never tuned for localization — sigma 2–8 sits directly on the 3.55–11.6 px lattice pitch, apparently the worst possible choice since the lattice is identical at every candidate. Suppressing it should have exposed the aperiodic content; instead accuracy **collapses to 0%** above the pitch (12.5/7.5/0.0/0.0/0.0% at sigma_lo 4/6/8/12/16). The lattice is what lets the correlation lock onto a position at all: registration and disambiguation want opposite filters, and the shipped (2.0, 8.0) is the best of twelve configurations. Finally, dense.py notes that selection among near-ties is *the periodic-residual ranking stage's job* — a stage the Phase 2 path never calls. Restoring it as a shortlist re-ranker reproduces the shipped result exactly (+1/–1, p=1.00), so it is not a missed opportunity either.

2. What worked: DoG band-pass, confirmed at n=280

results/phase2_experiments/exp01/summary.json, identical pose sweep and selection rule, only the ZNCC input differing:

Arm	≤1px	Rate	95% CI	Mean credit	Pts/40	Median s/pair
Raw	52/280	18.6%	[14.5,23.5]	0.189	7.5	3.49
DoG	83/280	29.6%	[24.6,35.2]	0.334	13.3	3.54

Paired: fixed 49, broke 18, net **+31**, McNemar **p=0.0002**. Gain concentrates where Set B's 0.55 weight sits — a low-frequency charging-drift suppression effect, not generic contrast (robust_contrast is affine and provably cannot move ZNCC, and did not).

Severity	Raw	DoG	Δ (pp)
0	26.4%	35.8%	+9.4
1	27.6%	28.9%	+1.3
2	15.9%	26.1%	+10.1
3	7.3%	29.3%	+22.0

Severity 3 quadruples but remains the floor in absolute terms (29.3% vs 35.8% at severity 0) — the hardest regime is improved, not solved.

Real example (results/phase2_failures/rank003_p2_val-000123_err466.7px.png): true site (238.29, 496.96), FinFET, severity 3; raw-input error 465.7 px, DoG-input 497.7 px — one of the 18 pairs DoG *breaks*, so the net +31 is not uniform. Half of measured misses land near an integer lattice translate of the truth with scale and rotation still correct: the matcher finds a period-consistent site, just not the right one.

Compact scoreboard

Block	Metric	Value	Evidence
Localization (n=280)	mean credit, DoG vs raw	0.334 / 0.189	exp01/summary.json
Pose localized (n=25)	scale / rotation credit	1.000 / 0.956–1.000	PHASE2_FINDINGS.md §1
Rejection (lockbox n=51)	F1, AUC, CI90	0.9024, 0.8707, [0.838,0.953]	presence_hgb.metadata.json
Confidence (lockbox n=74)	AUC vs raw ZNCC	0.892 vs 0.760	exp18_correctness.json
Runtime (n=60, uncontended)	median/P95/max	4.33/4.51/4.62 s	uncontended_runtime.json

3. What we killed, and why (do not re-run)

Hypothesis	Verdict	Evidence
Finer pose grid (45→99→187 poses)	no effect	identical credit to 3 decimals
5 anti-alias filters	no effect	byte-identical site on 24/25 pairs
Candidate pool caps (400, 32)	harmful	each silently deleted the true site
Lattice-normalized DoG bandwidth	harmful	–16 pp; measured pitches (3.55–11.6px) 1.8–22.5× too narrow
Nearest-centre rule on full surface	harmful	0/60 fixed at any margin; likely a scope bug (Phase 1 applies it inside a ranked pool, not ~800k raw positions)
Ranker retraining on Phase 2 corpus	no gain	–5.8 pp selection-given-pool (p=0.51), 8.5× cost
Lattice-sibling hard negatives	no gain	–3.8 pp vs incumbent
Bias/gain photometric compensation	harmful	AUC inverts to 0.429 at severity 3 (this session)

4. Final shipped architecture

Phase 1's periodic-aware ZNCC pipeline, **extended, not replaced**: dense full-resolution pose sweep (45 poses) over the disclosed scale/rotation ranges, DoG-band-passed ZNCC as the matching signal, refinement to subpixel/sub-degree credit, and **two separate fitted models**: a 17-feature HGB presence model for found (F1 0.90) and a 20-feature logistic correctness model for score (AUC 0.892 vs 0.760 for raw ZNCC). They answer different questions — *is the reference here* versus *is the coordinate I am reporting correct* — and a present-but-mislocalized pair must score low on the second while being 1 on the first, so one probability cannot serve both. See README.md “Phase 1 to Phase 2: what changed”.

5. Honest limitations

- **Selection is the only bottleneck that matters**: pool 90–95%, selection 15–43%. Harvest engineering will not move the score; new discriminating evidence is required and does not yet exist.
- **FinFET is not weaker than DRAM** under Phase 2 conditions — the Phase 1 README claim does not reproduce (25.0% vs 34.6% after band-pass, overlapping intervals on smaller per-architecture samples). This document supersedes that claim.
- **Rejection lockbox is small** (51 and 55 scenes, two slices). F1=0.9024 point estimate, but the 90% CI lower tail is **0.838**, so the F1≥0.90 bonus gate is at the boundary, not cleared.
- **Confidence AUC 0.892 (n=74) beats raw ZNCC 0.760** on the same lockbox, but n=74 is a small sample; the run is checkpointed to exp18_correctness.json.
- **Runtime is fine for the shipped path** — median **4.33s**, P95 4.51s, max 4.62s over 60 pairs measured uncontended with threads pinned to 4, so 4.3× headroom on the 20s cap and 0/60 over 5s. The figure rose from 3.54s when the correctness model was added. Forcing the budget down confirms the Deadline path sheds refinement and presence rather than overrunning, and the output contract holds at every budget tested to 0.5s. Separately, the heavier harvester used to measure 90–95% pool recall is not what ships — at that config 60/316 pairs (19%) exceeded the 20s cap (gt_pose_ceiling.json). Pool recall has not been re-measured on the exact shipped config.
- Pose credit (n=25) is a small sample; treat 1.000/0.956 as encouraging, not certain.
- All scored evidence is synthetic and self-generated; no organizer sample pairs were available at analysis time.

6. Next steps, ranked by evidence

- **Do not spend further effort on candidate ranking**. §1c bounds every such rule at ~35% even with perfect pose, and new features over the same surfaces inherit that bound. The only escape is a different similarity measure, and a materially different method is a no-appeal disqualification — closed by rule, not just by budget.
- Grow the rejection/confidence lockbox past 51–74 scenes before trusting the F1≥0.90 gate or the 0.892 AUC estimate.

7. Reproducibility

Python 3.11, CPU-only, no network at run time (scripts/verify_offline.py, audit-hook based). Weights ship inside the package (presence_hgb.pkl, correctness_lr.pkl, hgb_r2.joblib, SHA256-pinned in their metadata files) and are loaded script-relative. scripts/build_submission.py builds the zip from an explicit allow-list, fails if any required file is absent, then extracts the finished archive to a scratch directory and runs the documented entry point inside it. Validation is deterministic: 12 gates in scripts/validate_phase2.py, byte-identical regeneration, and results/phase2_experiments/exp01/per_pair.csv reproduces the DoG-vs-raw comparison in Section 2 row-for-row.